Band-1 Hadamard Order 7108 — Miyamoto (Double Paley-I H_1776 + C_1778) Miyamoto-only (NOT in SPK); Gap 2052 Still OPEN

H_888 Paley-I PASS; H_1776 Double PASS; C_1778 PASS; H_7108 Miyamoto PASS; Paley-I/II/Kronecker FAIL → Miyamoto-only (NOT in SPK); gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

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Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-16_Band1_Hadamard_6884_Miyamoto`; `2026-09-15_Band1_Hadamard_6836_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_7108_miyamoto/`.
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Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 888, 1776, and 7108 only** (verified $\max|HH^\top - nI| = 0$), via Miyamoto with Double Paley-I seed H_{1776} = Sylvester-double(H_{888} = Paley-I($q = 887$)) and symmetric Paley conference C_{1778} over prime $q = 1777 \equiv 1 \pmod 4$. Classical SPK for 7108: Paley-I **FAIL**s ($q = 7107 = 3 \cdot 23 \cdot 103$ composite); Paley-II **FAIL**s ($q = 3553 = 11 \cdot 17 \cdot 19$ composite); Kronecker **FAIL**s (4×1777). Therefore order **7108 is a Miyamoto-only bonus (NOT in SPK)** — same honesty class as 6836/6788. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior ship: **6884 IN SPK via Paley-I q=6883**. Prior SPK seeds **860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / 1668 / 6628 / 1656 / 1620 / 1608 / 6436 / 212 / 424 / 1696** retained; Miyamoto-only bonuses **6676/6484/6404/6388/6196/6788/6836/7108/...** excluded; **888/1776** folded into ONBOX when verified; **7108** NOT folded.

Abstract

Finite constructions of orders **888**, **1776**, and **7108**.

- H_{888} = Paley-I ($q = 887$) — Double half
- H_{1776} = Sylvester-double(H_{888}) — Miyamoto seed
- C_{1778} = Paley conference over prime $q = 1777$
- H_{7108} = Miyamoto(H_{1776}, C_{1778})
- Paley-I for 7108: **SKIP** ($q = 7107$ composite)
- Paley-II for 7108: **SKIP** ($q = 3553$ composite)

SPK: 7108 Miyamoto-only bonus (NOT in SPK). Seeds 888 / 1776 enter SPK. Prior SPK **860 / 1720 / 6884 / 1708 / 840 / 1680 / 6724 / ...** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_7108.py`

Results. overall_pass=true; units 106/106; wall 287.5642s; all max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 7108 = 4 \cdot 1777$; 1777 prime $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1776} and C_{1778} (Paley conference over prime $q = 1777$).
- Seed: Paley-I $q = 887$ (prime $\equiv 3 \pmod 4$) → H_{888} → Sylvester-double → H_{1776}.
- SPK: Paley-I $q = 7107$ **FAIL** (composite); Paley-II $q = 3553$ **FAIL**; Kronecker **FAIL** → **Miyamoto-only (NOT in SPK)**.
- Prior: 6884 IN SPK via Paley-I shipped; next optional was $q = 1777 \rightarrow n = 7108$.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_888 Paley-I	Paley-I q=887	0	0.747
H_1776 Double	Sylvester-double(H_888)	0	2.4032
C_1778	Paley conf q=1777	0	3.8394
H_7108 Miyamoto	Miyamoto(H_1776 Double, C_1778)	0	277.4532
H_7108 Paley-I	SKIP q=7107 composite $3 \cdot 23 \cdot 103$	—	—
H_7108 Paley-II	SKIP q=3553 composite $11 \cdot 17 \cdot 19$	—	—
H_7108 Kronecker	SKIP $4 \cdot 1777$ not admissible	—	—

SPK closure note

Seeds include prior SPK ships through **6884** lineage (**860/1720/6884** Paley-I/Double; **1708** Paley-II; **6724** Paley-II; **840/1680**; **6628**; **1668**; **1656**; **1620**; **1608**; **6436**; **212/424/1696**; **NOT** 6676/6484/6404/6388/6196/6788/6836/7108) + verified **888** / **1776** (Paley-I / Double); **exclude** Miyamoto-only bonuses including **7108**. Known count 1422. Next constructive gap **2052** (OPEN).

Quantity	Value
7108 in SPK	**False** (Miyamoto-only; excluded)
1776 in SPK	**True** (Double Paley-I)
888 in SPK	**True** (Paley-I q=887)
860 / 1720 / 6884 in SPK	**True** (prior)
1708 / 840 / 1680 / 6724 in SPK	**True** (prior)
6836 in SPK	**False** (Miyamoto-only; excluded)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Non-claims

No full Hadamard conjecture · no Clay-closed · no finished-ToE · no RH / P!=NP / YM / Frankl / NS-solved / AC-solved · gap 2052 remains OPEN · 7108 not claimed in SPK.

Repro

Lab: `/workspace/experiments/band1/hadamard_7108_miyamoto/`
Artifacts: `H888_paleyI_pm1.npz`, `H1776_double_pm1.npz`, `C1778_pm1.npz`, `H7108_miyamoto_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.